

The Effective Number of Parameters in Kernel Density Estimation

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- **KDE:** basic smoother of empirical density
- **Goal:** how to define an oracle measure of **effective degrees of freedom (EDoF)**?
- **Existing Methods:** all non-oracle,
 - Loader (1999): in context of local likelihood (two versions)
 - McCloud & Parmeter (2020): formulate KDE as regression problem
- **New Method:** gives oracle and non-oracle versions,
 - approximate empirical density and KDE with two systems of orthogonal polynomials, and relate coefficients with a smoothing matrix S
 - $\text{EDoF} \approx \text{Tr}(S)$
- **Connections:** with existing non-oracle methods
- **Example:** explicit calculations in Gaussian case
- **Bandwidth selection:** can use EDoF in AIC-like formulation?

$F(x)$ and $f(x)$: cdf and pdf of the continuous random variable X , supported on compact interval $[a, b]$.

- Oracle “Estimate”:

$$f_K(y) = \int K(x, y)f(x)dx = \mathbb{E}K(X, y).$$

- Plug-in estimate (KDE): uses empirical pdf $\rho(x)$

$$\hat{f}(y) = \int K(x, y)\rho(x)dx = \frac{1}{n} \sum_{i=1}^n K(x_i, y)$$

- Kernel $K(x, y) \equiv K_h(x, y)$ satisfies two essential properties:

P1: *Normalization.* $\int K(x, y)dy = 1$ for every x

P2: *Finite displacement.* for any function $g(x)$ supported on finite interval $[a, b]$, $g_K(y) \equiv \int K(x, y)g(x)dx$ is supported on interval $[\tilde{a}, \tilde{b}]$ which is also finite

- OPS $\{P_j\}$ is orthonormal with weight function $f(x)$:

$$\int_a^b P_j(x) P_k(x) f(x) dx = \delta_{jk}$$

- Expanding ratio $\rho(x)/f(x)$ implies:

$$\rho(x) = \frac{1}{n} \sum_{i=1}^n \delta(x - x_i) = \left(\sum_{k=0}^{\infty} c_k P_k(x) \right) f(x)$$

- Now expand $\hat{f}(y)/f_K(y)$ in terms of OPS $\{Q_j\}$ with weight $f_K(y)$:

$$\hat{f}(y) = \left(\sum_{k=0}^{\infty} b_k Q_k(y) \right) f_K(y)$$

- Relationship between coefficients b_k and c_k :

$$b_j = \int_{\tilde{a}}^{\tilde{b}} Q_j(y) \hat{f}(y) dy = \sum_{k=0}^{\infty} c_k s_{jk}$$

- The s_{jk} are elements of *kernel sensitivity matrix* S :

$$s_{jk} = \int_a^b \int_{\tilde{a}}^{\tilde{b}} K(x, y) Q_j(y) P_k(x) f(x) dy dx.$$

- The (infinite) vectors of coefficients $\{c_j\}$ and $\{b_j\}$, representing $\rho(x)$ and $\hat{f}(y)$, are related via:

$$\vec{b} = S \vec{c}$$

- View OPS for $\rho(x)$ as collection of independent "noise sources", since:

$$\mathbb{E}(c_j) = 0, \quad \text{and} \quad \mathbb{Cov}(c_k, c_j) = \frac{1}{n}$$

- Now:** calculate EDoF by considering how these noise sources are suppressed by smoothing...
- Ye (1998), Hastie et al (2009): EDoF in general smoothing problem is proportional to covariance between fit and data:

$$\mathbb{Cov}(\vec{b}, \vec{c}) = \mathbb{Cov}(S\vec{c}, \vec{c}) = \frac{1}{n}S$$

- Fine-tune S by:
 - eliminating 1st row & column (which are zero): $S \mapsto \tilde{S}$
 - symmetrizing (ensures positive-definiteness)
 - taking trace (sums all individual contributions)
 - normalizing (scale by covariance of $\vec{c}$)

Viewing $\tilde{S}$ as a *smoothing* matrix, the trace formulation below also coincides with a definition for EDoF proposed by Buja, Hastie & Tibshirani (1989) for linear smoothers.

$$\text{EDoF} := \nu = \text{Tr}(\tilde{S}\tilde{S}^T) = \sum_{j,k=1}^{\infty} s_{jk}^2 = \sum_{j,k=0}^{\infty} s_{jk}^2 - 1$$

Theorem

- density $f(x)$: continuous, supported on compact interval $[a, b]$.
- kernel $K(x, y)$: satisfies P1 and P2.
- $\{P_j(x)\}$ and $\{Q_j(y)\}$: OPS systems w.r.t. weight functions $f(x)$ and $f_K(y)$.

Then, we have that:

$$\nu_h = \int_a^b \int_{\tilde{a}}^{\tilde{b}} \frac{K_h^2(x, y)}{f_K(y)} f(x) dy dx - 1.$$

- Use **plug-in** from closed-form expression:

$$\hat{\nu}_h = \frac{1}{n} \sum_{i=1}^n \int_{\tilde{a}}^{\tilde{b}} \frac{K_h^2(x_i, y)}{\hat{f}(y)} dy - 1$$

- Approx uncertainty of $\hat{\nu}_h$: take limit $n \rightarrow \infty$ (**for fixed h ...**)
- **Nonparametric Delta Method**: gives relevant $n \rightarrow \infty$ asymptotics

$$\sqrt{n}(\hat{\nu}_h - \nu_h) \xrightarrow{d} \mathcal{N}(0, \omega_h^2), \quad \omega_h^2 = \int_{\tilde{a}}^{\tilde{b}} \int_{\tilde{a}}^{\tilde{b}} \frac{\gamma(w, z)}{\mu_{1,h}(w)\mu_{1,h}(z)} dw dz,$$

- (Asymptotics for $n \rightarrow \infty$ & $h \rightarrow 0$: challenging...)

- **Loader (1999):** in context of **local likelihood** defines two kinds:

$$\hat{\nu}_1 = \sum_{i=1}^n \text{infl}(x_i), \quad \text{and} \quad \hat{\nu}_2 = \sum_{i=1}^n \text{vari}(x_i).$$

- (**Note:** degree zero local likelihood $\iff$ KDE)
- **McCloud & Parmeter (2020):** express KDE as a linear **regression-like** object, propose:

$$\hat{\nu}_3 = \frac{1}{n} \sum_{i=1}^n \frac{K_h(x_i, x_i)}{\hat{f}(x_i)}$$

- Using 1st order Taylor expansion of $\log \hat{f}_{-i}(x_i)$ about $\log \hat{f}(x_i)$, we show that:

$$\hat{\nu}_1 \approx \hat{\nu}_3$$

(Loader 1 is approx equivalent to McCloud & Parmeter)

- Since $\text{vari}(x_i) = \mathbb{V} \log \hat{f}(x_i)$, we show that Delta Method gives:

$$\hat{\nu}_2 \approx \hat{\nu}$$

(Loader 2 is approx equivalent to our EDoF)

But: we have an ORACLE formulation! (There is no ν_1 & $\nu_2 \dots$)

- $f(x) \sim \mathcal{N}(0, \sigma^2)$,
- $K_h(x, y) = \frac{1}{h} K\left(\frac{y-x}{h}\right) \sim \mathcal{N}(0, h^2)$
- Hermite OPS's $\{P_j\}$ and $\{Q_j\}$ (differ by scaling)
- $f_K(y) \sim \mathcal{N}(0, \tau^2)$, where $\tau^2 = \sigma^2 + h^2$
- Leads to tractable integral for elements of S (diagonal!):

$$\begin{aligned} s_{jk} &= \frac{1}{2\pi h\sigma} \int_{\mathbb{R}^2} P_j(y, \tau) P_k(x, \sigma) \exp\left\{-\frac{x^2}{2\sigma^2} - \frac{(x-y)^2}{2h^2}\right\} dx dy \\ &= \left(1 + \frac{h^2}{\sigma^2}\right)^{-j/2} \delta_{jk} \end{aligned}$$

- Leads to EDoF:

$$\nu = \frac{\sigma^2}{h^2} \equiv \nu_G$$

- **Model log-likelihood:**

$$\ell_0(h) = \sum_{i=1}^n \log f_K(x_i|h)$$

- **Empirical log-likelihood:**

$$\ell(h) = \sum_{i=1}^n \log \hat{f}(x_i|h)$$

- Kullback–Leibler distance $f_K(x|h) \mapsto f(x)$ (model $\mapsto$ truth):

$$\text{KL distance} = \text{constant} - \mathbb{E} \log f_K(X|h),$$

- Thus: **Target** to maximize (for optimizing h):

$$\text{Target} = \mathbb{E} \ell_0(h).$$

But: we don't have access to $\ell_0(h)$...

Solution: use $\ell(h)$

- Taylor expansion of $\log \hat{f}(x|h)$ to $O(n^{-1/2})$ gives:

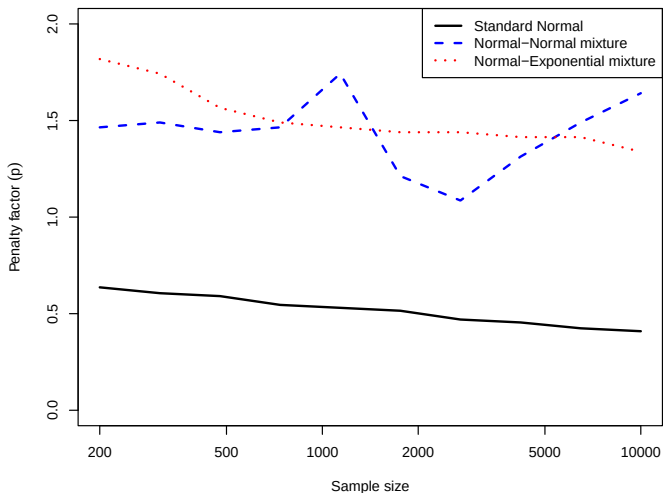
$$\begin{aligned}\mathbb{E}\ell(h) &\approx \mathbb{E}\ell_0(h) + \frac{\sigma^2}{h^2} \\ &= \text{Target} + \nu_G\end{aligned}$$

- Thus a (1st order) unbiased estimate of KL is AIC-like quantity:

$$AIC(h) = \ell(h) - \nu_G(h)$$

(Optimal penalty factor is 1, just like usual AIC...)

What is Optimal Penalty Factor in AIC? Examples



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Thank You!